



Capability Assessment

Request for Information (RFI)

Joint Simulation Environment (JSE) Operations and Support (O&S)

12 March 2026

Prepared By:

**Air Force Lifecycle Management Center
Training Directorate (AFLCMC/WN)
Advanced Training Capabilities Division (AFLCMC/WNR)
1895 5th Street, Building 46
Wright-Patterson AFB, Ohio 45433**

THIS PAGE INTENTIONALLY LEFT BLANK

Table of Contents

1. REQUEST FOR INFORMATION (RFI) FOR PLANNING PURPOSES	2
2. PROGRAM REQUIREMENTS	3
2.1 Joint Simulation Environment (JSE) Program.....	3
2.2 Background	3
2.3 System Description	4
2.3.1 JSE Operational View (OV-1)	4
2.3.2 A JSE site operation is comprised of six building blocks:.....	5
2.3.3 JSE Technical Baseline (JTB)	5
2.3.4 JSE Platform Package (JPP)	6
2.3.5 Blue Force Platforms	6
2.3.6 Red Force Platforms	7
2.3.7 Distributed Interactive Simulation (DIS) Gateway.....	7
2.4 Operational Locations and Operational Capability Phasing	7
2.4.1 Joint Integrated Test and Training Center–Nellis AFB, NV (JITTC-N)	7
2.4.2 Joint Integrated Test and Training Center –Joint Base Elmendorf- Richardson, AK (JITTC-E).....	9
2.5 O&S Requirements	10
2.5.1 Personnel Security	11
2.5.2 Information/Industrial Security	11
2.5.3 Contractor Facility Requirements	11
2.6 Contract Specifications	11
2.7 Capability Assessment/RFI Response	11
2.8 General Information.....	12
3. CAPABILITY INFORMATION	13
3.1 JSE Operations and Support Capability Information.....	13

1. REQUEST FOR INFORMATION (RFI) FOR PLANNING PURPOSES

This RFI/Capability Assessment is in accordance with (IAW) Federal Acquisition Regulation (FAR) 52.215-3 – Request for Information or Solicitation for Planning Purposes (Oct 1997). Responses to this notice are not offers and cannot be accepted by the Government to form a binding contract.

The Government does not intend to award a contract based on this RFI or to otherwise pay for the information solicited except as an allowable cost under other contracts as provided in subsection 31.205-18, Bid and Proposal costs, of the FAR. Companies responding to this RFI/Capability Assessment are advised that participation does not ensure involvement in future solicitations or contract awards. The Government will not reimburse any company or individual for any expenses associated with preparation or participation in this RFI.

The acquisition strategy has not yet been determined. Sources Sought results will assist the Air Force in determining whether these requirements will be an unrestricted competition or a small business set-aside task order as well as if it will be awarded under Training Systems Acquisition (TSA) IV or if another contract vehicle will be used. The Government is assessing the capability of companies that may possess the expertise to competently and competitively respond to the following requirements for the **Joint Simulation Environment Operations & Support** to provide:

- ☒ Program Management/Systems Engineering
- ☐ Upgrades and Modifications (e.g., Concurrency Upgrades, Training Enhancements, etc.)
- ☒ Training Systems Support (e.g., Contractor Logistics Support [CLS], Training System Support Center [TSSC], etc.)
- ☒ Instructional Services
- ☐ Products
- ☐ Design Concept and Prototype Development
- ☒ Interoperability
- ☒ Qualification and Test

Although “proposal” and “offeror” may be used in this RFI, your response will be treated as information only. It shall not be used as a proposal.

2. PROGRAM REQUIREMENTS

2.1 Joint Simulation Environment (JSE) Program

For the JSE Program, the contractor shall possess the capabilities to fulfill training operations and product support functions at **Joint Integrated Test and Training Centers (JITTCs)**. The JSE Program contains two JITTCs: Joint Integrated Test and Training Center–Nellis AFB, NV (JITTC-N) and Joint Integrated Test and Training Center –Joint Base Elmendorf-Richardson, AK (JITTC-E). Additional sites may be included as part of this solicitation. These training sites will:

- Contain multiple training systems (JSE-compliant and non-JSE-compliant) developed by other airborne platform program offices and their Original Equipment Manufacturers (OEMs)
- Be supported by many COTS software and hardware components
- Contain server farms hosting JSE software components along with the cockpits support racks associated with the training systems
- Use complex audio/visual systems for mission brief and debrief
- Operate on secured networks supporting Secret/Collateral, TS/SCI, and SAP information.

2.2 Background

The JSE is an ACAT I-equivalent Software Acquisition Pathway (SWP) program. The US Navy (USN) originally developed this modeling and simulation (M&S) capability to conduct F-35 Initial Operational Test & Evaluation (IOT&E) in a virtual environment. Recognizing JSE's potential, the AF established the program of record to continue developing this capability into a digital test and training range for 4th, 5th, and 6th generation aircraft in partnership with the USN. The first goal of this acquisition is continuous integration/continuous delivery (CI/CD) of the core software capabilities known as the JSE Technical Baseline (JTB). This includes producing physical environment and threat models that enhance the fidelity of the virtual battlespace required for aircraft testing and training. The second goal of this program is simulator training systems integration. The JSE program is collaborating with aircraft program offices and their aircraft OEMs to develop and integrate Operational Flight Program (OFP)-based, operator-in-the-loop simulator solutions using high fidelity models of aircraft mission systems and sensors with the JTB. The third goal is to field and sustain the JSE capability at designated sites for aircraft test and training operations. Successful execution of this program will deliver to the warfighter the ability to train for the next conflict in a secure virtual battlespace.

2.3 System Description

JSE provides a government owned, physics-based computer environment capable of simultaneous interactions between manned aircraft simulators and thousands of virtual friendly and enemy air and surface entities. When hosted on appropriate battlespace hardware at a government site and paired with high-fidelity operator-in-the-loop OFP-based simulators, JSE allows the warfighter to execute testing, training, or tactics development and validation to an unprecedented level of fidelity. To quote pilot feedback received regarding current JSE capabilities, “The density and realism of training experienced with JSE enables the warfighter to learn and improve at a pace unachievable anywhere else.”

Further JSE development and integration will enable simultaneous support to multiple airborne platforms. The JTB, which consists of integrated software/system components, is hosted on commercial-off-the-shelf (COTS) hardware, leverages third-party COTS software where necessary, and operates within a closed network with operator-in-the-loop simulators. A Government Simulation Interface (GSI) provides the standards for developing and integrating operator-in-the-loop capabilities for system under test and training along with other mission players—both friendly and hostile. The software is designed to be scalable, flexible, adaptable, modular, and configurable to support customers’ needs. Lastly, The JSE architecture, environment, and overall capabilities will be enhanced to meet new user capability requirements.

2.3.1 JSE Operational View (OV-1)

Error! Reference source not found. is a simplified Operational View (OV-1) representing the systems architecture of the JSE fielded configuration. The JTB located in the center of the diagram provides the synthetic environment, modeled threats for integration with virtual blue (friendly) and red (enemy) airborne platform simulators, scenario generation, operational control, and brief/debrief functions. Interoperable with the JTB, Blue Forces Platforms using Platform-in-a-Box (PIAB) software and Red Force Platforms using Virtual Air Threats software represent the operator-in-the-loop simulators flying in the virtual environment. These software components are built on a solid foundation of existing DoD M&S technologies.

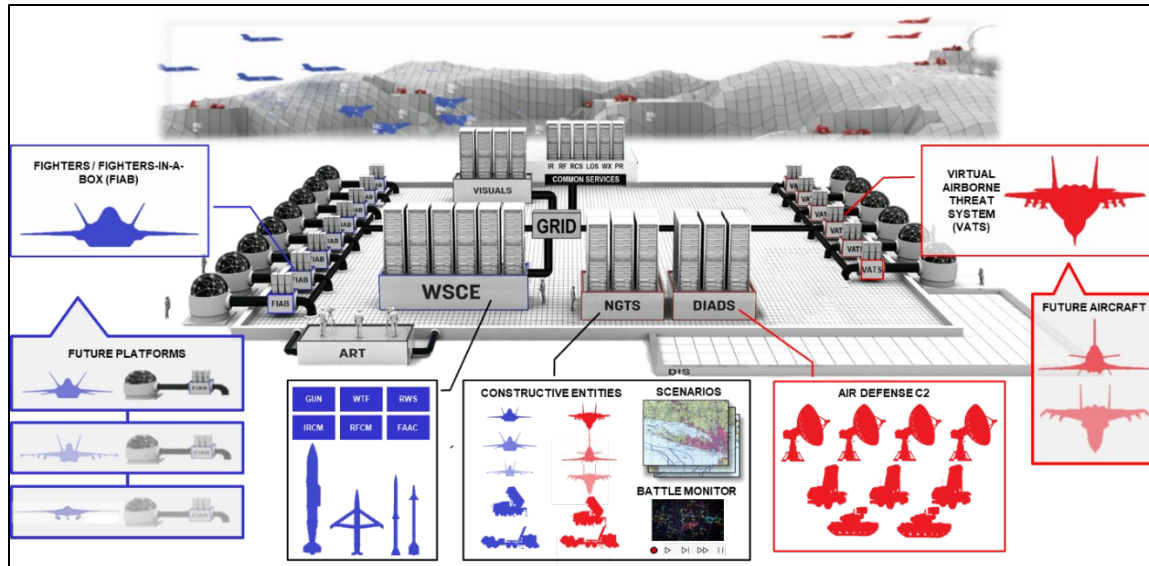


Figure 2.1. JSE OV-1 (Simplified)

2.3.2 A JSE site operation is comprised of six building blocks:

- JTB software that provides a virtual battlespace environment that is extensible and modular.
- COTS physical computing infrastructure (e.g., racks, servers, data storage, routers, and peripherals), which implements the battlespace within a closed network.
- Software providing integrated operator-in-the-loop airborne platforms—high fidelity OFP-based representation of the manned aircraft or weapons system operator station in simulation, including mission systems and sensors, for Blue and Red Force Platforms
- High, medium, or low fidelity pilot vehicle interfaces including cockpit representations, audio, and visual display systems.
- Control rooms with operator stations and briefing rooms with playback capabilities that enable mission planning, execution, and debrief.
- A networked facility that securely contains all the above and manpower to operate it.

2.3.3 JSE Technical Baseline (JTB)

The JTB is the integrated suite of high-fidelity, government-owned software components listed below, which produces a common modeling and simulation environment for airborne platform test and training. The architecture includes services, simulation messaging, and operator interface systems allowing scenario development, pre/post brief, mission execution, brief/debrief operations, white-carding, and integration of operator-in-the-loop simulator solutions. Naval Air

Warfare Center Aircraft Division's (NAWCAD) Integrated Battlespace Simulation and Test (IBST) Department manages JTB development, integration and release configuration.

Global Reusable Interface Domain (GRID), a core JTB component, is a high-fidelity network library with a fast message passing simulation protocol that provides physics-based interaction and propagation effects between all constructive and virtual entities in a simulated environment. Engineered to support large force exercises of varying complexities, GRID can scale horizontally by adding computer hardware resources. GRID provides simulated environment services: Entity, Infrared, Radio Frequency, Radar Cross Section, Denial Zones, Environment, Endgame/Probability of Removal, Line of Sight, Laser. Future services include enhanced weather and datalinks. GRID also includes a DIS (Distributed Interactive Simulation) Gateway for interfacing with the Digital Integrated Air Defense System (DIADS), specific Next Generation Threat System (NGTS) entities, Weapon Server Common Environment (WSCE), and systems under development.

2.3.4 JSE Platform Package (JPP)

JPP software is organically developed to facilitate the interface between the JTB and the Blue Force Platforms. JPPs are designed to account for the unique features of each Blue Force Platform connecting to the virtual battlespace environment.

2.3.5 Blue Force Platforms

Although JSE was initially deployed at NAS Patuxent River, MD to test and train with F-35 aircraft simulators, the capability will expand to incorporate other weapon system platforms from across a multi-service kill chain in large scale scenarios (e.g. Space Command and Control, Air Battle Management, ground fires). The JSE PO will partner with weapon system program offices to develop Blue Force Platforms solutions that are interoperable with each other and JSE. Blue Force Platforms include operator-in-the-loop solutions for pilots and C4ISR/Electronic Warfare Combat Systems Officers. Depending on what is required at each site, these systems could be low, medium, or high-fidelity simulator solutions. Blue Force Platforms comprise three subcomponents as shown in Error! Reference source not found..

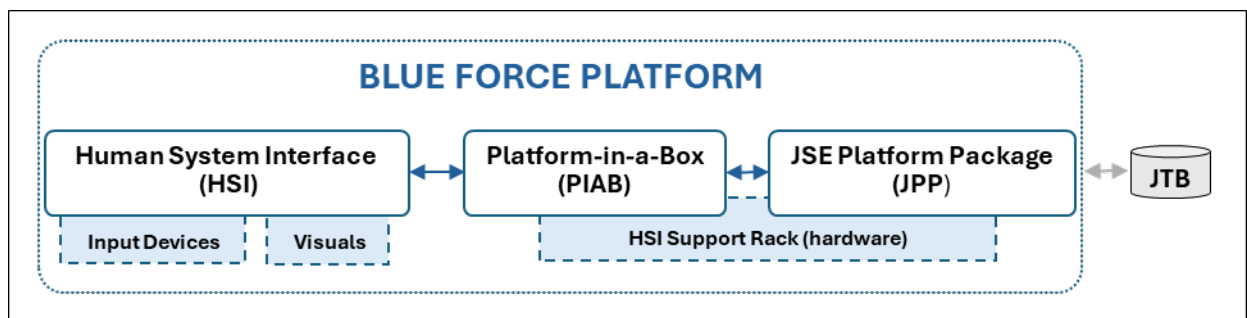


Figure 2.2. Blue Force Platform Subsystems

2.3.6 Red Force Platforms

Within the synthetic environment, red threats will include both constructive (computer-driven) and virtual (operator-in-the-loop) entities. Virtual Red Force Platforms comprise the Virtual Air Threat Systems (VATS), JPP, and HSI.

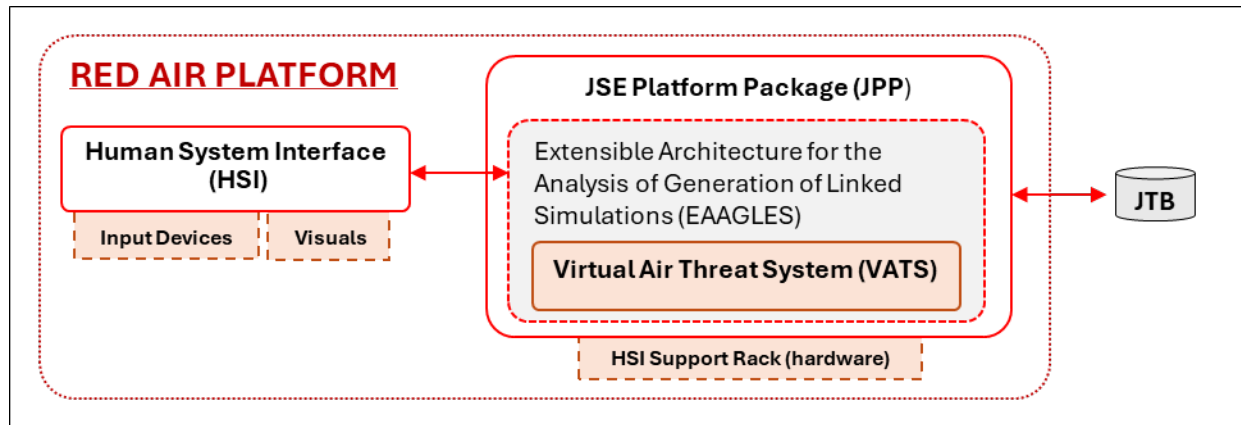


Figure 2.1. Red Air Platform Sub-Systems

2.3.7 Distributed Interactive Simulation (DIS) Gateway

As part of GRID services, DIS is an IEEE standard for conducting real-time platform-level wargaming across multiple host computers. It is used worldwide by military organizations and other organizations involved in space exploration and medicine. DIS is used to communicate various state messages between GRID and other JTB components such as NGTS, WSCE, and Event Recording services. To support JITTC-N current operations, DIS is also being used to develop interfaces between JSE and non-complaint simulator training systems (e.g., legacy F-15E training device) until their PIAB solutions are developed and deployed.

2.4 Operational Locations and Operational Capability Phasing

Currently, the Air Force is planning for at least two major JSE training sites. The JSE Program will be acquiring capabilities to support operations and sustainment at these sites. US Navy training sites are being considered for inclusion in the solicitation.

2.4.1 Joint Integrated Test and Training Center–Nellis AFB, NV (JITTC-N)

The USAF Warfare Center's 31st Combat Training Squadron (31 CTS) is converting four existing facilities to serve as the AF's first JITTC supporting HEAT3 mission development and execution. This is in support of AF, Joint, and Coalition partner nation warfighters to deliver full spectrum, multi-domain combat readiness to AF Weapons School syllabus events and exercises, COCOM exercises, OPLAN rehearsals, Flag-level exercises, and operational test risk reduction. JITTC-N currently possesses a JSE capability including 8x F-35 Blue Force Platforms using

medium fidelity cockpits. This capability may be reconfigured to operate as 4x F-35 and 4x Virtual Air Threat Solution (VATS) adversaries for Blue versus manned Red Air scenarios.

The 31 CTS will support continuous verification of Blue Force Platforms integrated with JSE in coordination with the 445th Test Squadron and 29th Test and Evaluation Squadron. Furthermore, a small development lab will be installed at this site to perform model development within the program's Collaborative Development Environment (CDE) known as the JSE Crucible.

The 31 CTS will have a Concept of Operations (CONOPS) (pending HQ ACC approval) describing the operations tempo and phased expansion of JITTC-N with JSE-compatible Blue Force Platforms to support simultaneous training events. The standalone networked campus is projected to have 61 Blue/Red Force Platforms (fighters and bombers) and 82 C2ISR Battlestations by FY31. 44 Blue/Red Force Platforms and 14 C2ISR Battlestations are projected to be in place when this solicitation is awarded in FY28. On-premises office desk space and equipment storage space will be limited at this location. Quantities and installation schedules of these training systems are subject to change based on system development and facility renovation timelines.

Training Systems	FY28	Post-FY28	Description
F-35	16	0	Medium fidelity dome/cockpit/CSR
F-22	8	0	Medium fidelity dome/cockpit/CSR
VATS (med)	8	0	Medium fidelity dome/cockpit
VATS (low)	12	0	Low fidelity desktop
F-15EX	0	2	Medium fidelity dome/cockpit/CSR
F-15E	0	4	Medium fidelity dome/cockpit/CSR
F-16	0	4	Medium fidelity dome/cockpit/CSR
B-52	0	2	Medium fidelity dome/cockpit/CSR
Other Bomber Aircraft	0	2	TBD
C2ISR Battlestations	14	69	Combat Systems Officers' control stations for EA-37B, E-7, RC-135

Figure 2.4. Planned JITTC-N Training Systems

2.4.2 Joint Integrated Test and Training Center –Joint Base Elmendorf-Richardson, AK (JITTC-E)

The 3rd Wing is constructing a single facility to serve as JITTC-E, which will support US and partner nation capabilities for test and training activities in the Pacific Air Forces (PACAF). The MILCON project is estimated for completion in FY29. Central to its concept of operations is OPLAN design, generation, rehearsal, and the flexibility and capacity to perform multi-domain training across the full spectrum of operations.

The 3rd Wing is developing a CONOPS describing the operations tempo and phased expansion of JITTC-E with JSE-compatible Blue Force Platforms. JITTC-E is projected to have an initial package of 16 Blue/Red Force Platforms one year after this solicitation is awarded in FY28. The full capability will consist of Fighter Agnostic Cockpits (FAC) to support Large Force Exercises (LFE). Development of the FAC solution will start this year. JITTC-E is projected to have 203 Blue/Red Force Platforms (including 76 FACs), and 107 C2ISR Battlestations by FY35. Quantities and installation schedules of these training systems are subject to change based on system development and facility construction timelines.

Site Components	Installed FY29	Installed Post FY29	Description
F-35	8	0	Medium fidelity dome/cockpit/CSR
F-22	4	0	Medium fidelity dome/cockpit/CSR
VATS (low)	4	0	Low fidelity desktop
VATS (med)	0	20	CSR and HOTAS only for FAC integration
F-35 (FAC)	0	32	CSR and HOTAS only for FAC integration
F-22 (FAC)	0	16	CSR and HOTAS only for FAC integration
F-47 (FAC)	0	12	CSR and HOTAS only for FAC integration
F-15EX (FAC)	0	12	CSR and HOTAS only for FAC integration
F-16 (FAC)	0	4	CSR and HOTAS only for FAC integration
B-52	0	2	Full training device
Other Bomber Aircraft	0	9	TBD

Site Components	Installed FY29	Installed Post FY29	Description
FAC	0	76	Single cockpit solution for all fighter aircraft
C2ISR Battlestations	0	51	Combat Systems Officers' control stations for EA-37B, E-7, RC-135
JSE Operator Stations	44	63	Ops cell functions for training events

Figure 2.5. Planned JITTC-E Training Systems

2.5 O&S Requirements

The contractor will be required to manage all aspects of the JITTC operations. This will include:

- Contract transition activities including transfer of responsibility from the current O&S contract
- Program management of all activities performed by contractor personnel
- Establishing processes that integrate all technical disciplines including engineering, configuration management, logistics management, and testing to operate and maintain the JITTCs
- Assisting the Government with training event planning and execution with the support of pilot instructors and white cell personnel
- IT systems administration and simulator technician responsibilities
- JTB software and hardware installation, integration, and acceptance testing
- Managing commercial-off-the-shelf (COTS) software license agreements
- Logistics/supply support including required hardware procurements for COTS hardware sparing and solutions, establishing Contractor-Inventory Control Points (C-ICP), and collecting Reliability, Availability, and Maintainability (RAM) data
- Packaging, handling, storage and transportation of JITTC equipment
- Cybersecurity compliance and Information Assurance (IA) management
- O&S of PIABs in coordination with the respective airborne platform program offices and their OEMs (Note: Platform program offices will be responsible for maintaining concurrency with the weapon system)

- Establishing Associate Contractor Agreements (ACAs) as directed by the Government or deemed appropriate to meet the O&S requirements

2.5.1 Personnel Security

The contractor will be required to possess a valid Secret Facility Clearance (FCL) and CAGE code indicating the company's level of clearance and the required security support to maintain the facility. Surge development may be required to occur at the contractor's facility due to space limitations within government facilities. The contractor will be required to possess the ability to send classified visit requests for visits to government facilities. The Defense Information System for Security (DISS) web application is the tool used for sending classified visits requests. Facility Security Officers (FSO) or the contractor's security administrators should have access to this system or have a sponsor with access.

2.5.2 Information/Industrial Security

The contractor must comply with the National Industrial Security Program (NISPOM), 32 CFR Part 117 and DoD Information Security Program, DoD 5200.01 parts 1-3, and DoD Instruction Controlled Unclassified Information (CUI) DoDI 5200.48.

2.5.3 Contractor Facility Requirements

On-premises office desk space and equipment storage space will be limited at JITTC-N. The contractor will have to plan for off-base office space and C-ICP.

2.6 Contract Specifications

The contract will commence on or about the 4th Quarter of CY27 to allow for up to a 6-month transition period for JITTC-N. The Government anticipates contract period of performance to be a 1-year base period with 4 1-year option periods. The Government anticipates the use of:

- Firm Fixed Price (FFP) Contract Line Item Numbers (CLINs): contract transition, program and engineering management, JITTC O&S, and JITTC facility modifications/upgrades
- Fixed Price Incentive Firm (FPIF) CLINs: Maintaining JITTC 95% operational availability threshold
- Cost Reimbursable CLINs: Travel and studies/analysis tasks

2.7 Capability Assessment/RFI Response

The results of this assessment will allow the Government to decide as to whether these efforts will be set aside for the Small Business or issued Full and Open. It is advised that your company provide a sufficient response to all capability questions to allow the Government to appropriately assess your company's capabilities.

If there is demonstrated interest and capability sufficiently identified among the small business prime contractors, then a Small Business Set-Aside will be considered. A key factor in determining whether an acquisition will be a Small Business Set-Aside is that two or more potential small business prime contractors must be capable of performing at least 50% of the effort.

2.8 General Information

In your response, please provide the following general information for your company/institution and for any teaming partners:

- If subcontracts or teaming are anticipated to be used, provide a reasonable expectation for both small and other than small business utilization as a percentage of total task order value. Provide specific details on how your company will comply with FAR clause 52.219-14, Limitations to Subcontracting. Delineate the small business utilization percentage and provide rationale for the recommended percentage. Teaming and/or subcontracting arrangements should be clearly stated, and previous experience in teaming must be provided.
- What data do you require from the Government to assemble a proposal for this effort?

Direct and concise responses are preferred. Marketing material is considered an insufficient response to this Capability Assessment/RFI and will not be used to assess capabilities. Please use the following file-naming convention: <<Company Name>> <<Program Name>> RFI Response. **Limit responses to electronic submissions no more than ten 8.5"x11" pages for all programs in total, using Word or PDF format.** This includes all graphs and charts. A point of contact that can represent your company should be identified to answer further Government inquiries and provide explanations that might be needed regarding the information provided. Respondents should indicate which portions, if any, of their response are proprietary and should mark them accordingly.

Formal responses should be submitted via electronic submission through email sent to Buyer Matthew Mercer, matthew.mercer.5@us.af.mil and Contracting Officer Elizabeth Moles, elizabeth.moles@us.af.mil

Further questions may be e-mailed to the Contracting Officer, elizabeth.moles@us.af.mil, Program Manager, john.kovacic@us.af.mil, and Logistics Management Specialist, gregory.zele@us.af.mil. **Responses are due NLT 11:59 PM EST, 10 April 2026.** The Government may converse with respondents regarding this RFI. Such conversations are intended to clarify potential capability to perform the requirements.

3. CAPABILITY INFORMATION

In response to this RFI, please provide the following cross-program and training system-specific capability information for your company and for any teaming partners. If a responding company is capable of some, but not all, of the consolidated requirements stated, please provide a response that explains the requirements for which you are capable.

3.1 JSE Operations and Support Capability Information

In response to this Capability Assessment/RFI, please provide the following capability information specific to the JSE Program requirements.

1. As the proposed effort is considered by the Government to be one for Services, small business offerors will be required to comply with subpara (c)(1) of FAR 52.219-14, which requires that “[a]t least 50 percent of the cost of contract performance incurred for personnel shall be expended for employees of the concern.” Please describe in detail how you will satisfy the requirements of FAR 52.219-14(c)(2) for this effort.
2. Training sites will contain multiple training systems (JSE-compliant and non-JSE-compliant systems) developed by multiple OEMs, be supported by many COTS software and hardware components from multiple vendors, house large supporting server farms, using complex audio/visual systems for mission brief/debrief, and operating on secured networks with Secret/Collateral, TS/SCI, and SAP information.
 - Describe your capabilities and experiences to operate and maintain these sites as described above including satisfying the software and hardware testing, integration, installation and training planning/execution.
 - Describe any existing Associate Contract Agreements or the ability to establish ACA with other companies that either produce or support Air Force training systems and subcomponents.
 - Describe your capabilities and experiences working in SCIF and SAPF areas and ensuring the training operations comply with security requirements (i.e., industrial, physical, personnel, operations).
3. Describe your capability and experiences to satisfy facility engineering which supports training equipment installations and classroom modifications in accordance with SWAP-C requirements.
4. Post contract award, the two training sites described above may be activated within a 12-24 month period. At the time of contract and site activation, these training sites are forecasted to primarily possess initial operating capabilities consisting of F-22, F-35, and Red Force training

systems. In the years following, additional JSE-compliant airborne platform training systems will be delivered to these sites to reach the desired Full Operational Capability.

- Describe your ability to execute contractor transition for an existing training site at Nellis AFB while also activating a new training site at Joint-Base Elmendorf-Richardson, AK within a 12-24 month period based on the training capabilities projected to be in place in 2028-2029.
 - Describe your capability to simultaneously satisfy manpower and training support requirements for the two training sites.
 - Describe your ability to increase O&S capabilities (e.g., manpower, C-ICP) commensurate with the fielding of additional training systems at the two JSE training sites.
5. Describe your capabilities/systems to obtain OEM technical data.
 6. Describe your ability to facilitate training operation including training scenario development, blue and red force pilots, white cell, event execution, and mission brief/debrief.
 7. Describe your capabilities and experience with Contractor-Inventory Control Point (C-ICP):
 - Establishing and executing a C-ICP
 - If you have established a C-ICP, where is it located and how long did it take to build and prepare the facility for warehousing?
 - Managing government furnished property (GFP) and, if so, how many line items of GFP have you managed?
 - Do you currently have access to a Defense Logistics Management Standards (DLMS) compliant inventory management system?
 - If you currently have a DLMS-compliant inventory management system, are you actively interfacing with the government Accountable Property System of Record (APSR) Enterprise Logistics Management System (ELMS)?
 8. Have any of your contracts required the submittal of DI-MGMT-80259 Physical Inventories Report (commonly referred to as the Chief Financial Officer report)? If yes, how many approved submittals have you completed?
 9. Describe the major risks or issues you foresee in executing an effort of this scope. What steps will you likely take to mitigate these risks?